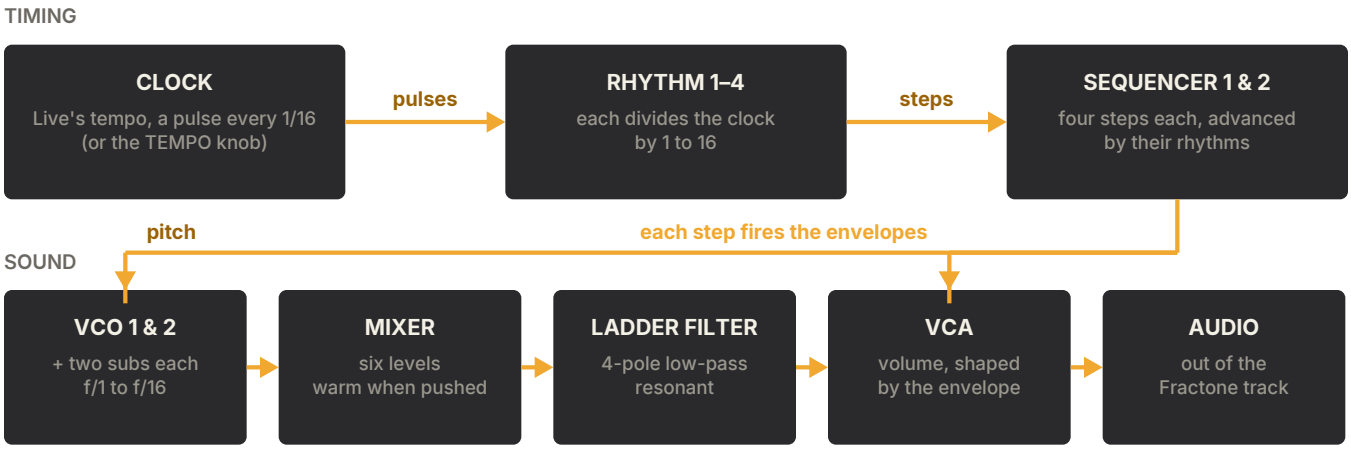
FRACTONE

f/1 to f/16: the undertone series

Getting started: your first polyrhythm in Ableton Live

Fractone is a polyrhythmic undertone synthesizer. Two oscillators each carry two subharmonics, the note divided by 1 to 16, and four rhythm generators divide one clock by 1 to 16 to drive two four-step sequencers. It makes its own sound, so one MIDI track in Live is all it needs. This guide gets it playing in a couple of minutes, then builds your first polyrhythm one step at a time.



How a note happens: the clock drives the rhythms, the rhythms advance the sequencers, and the sequencers set the pitch and fire the envelopes.

You'll need

- Ableton Live 11 or 12, any edition that can load VST3 or Audio Unit plug-ins.
- Fractone installed. Builds are on the GitHub releases page; on macOS **Fractone.vst3** goes in **~/Library/Audio/Plug-Ins/VST3** and **Fractone.component** in **~/Library/Audio/Plug-Ins/Components**. Building from source installs both for you.
- Nothing else: no second track and no other instrument.

> **Coming from Dicey?** Fractone works the other way round. It makes sound and listens to MIDI, so there's no routing to set up: the track it sits on is the track you hear.

Set up

1 Let Live see Fractone

Open **Live → Settings** (⌘,) and choose the **Plug-Ins** tab. Turn on **Use VST3 Plug-in System Folders** (or **Use Audio Units**), then press **Rescan**.

In the browser, open **Plug-Ins**. Fractone is under **Taddei**.

2 Put it on a MIDI track

Insert a MIDI track (**Create → Insert MIDI Track**, or ⌘⇧T) and drag **Fractone** onto it. It opens on the **Init** patch: VCO 1 and its first subharmonic, played by Sequencer 1.

3 Press play

Fractone follows Live's tempo and transport. Press Live's play button and you'll hear a four-note phrase, while the step LEDs walk across Sequencer 1. The status display on the bottom rail reads **SYNC 120.0 BPM 1/16 RUNNING**.

Stop Live and Fractone stops too. Starting again, jumping or looping resets the sequencers and rhythms, so every pass begins on the same downbeat.

> **Nothing?** The display says **WAITING FOR HOST** until Live is playing. If it says **PLAY OFF**, click **PLAY** on the panel. Troubleshooting on the last page has the full list.

How the clock works

Each clock pulse is a 1/16 of Live's tempo (**RATE** on the bottom rail changes it, from 1/1 to 1/64). A rhythm generator counts those pulses: Rhythm 1 in the Init patch is set to 1/2, so it fires on every second pulse and Sequencer 1 moves in eighth notes. Turn the RHYTHM knob clockwise for faster, counter-clockwise for slower; hover it to read the division.

Turn **SYNC** off to run from the panel's own **TEMPO** knob instead, one pulse per beat from 20 to 3000 BPM, started and stopped with PLAY.

Your first polyrhythm

Start from Init, with Live playing. Each step adds one idea.

1 Tune the steps

Turn Sequencer 1's **STEP** knobs. Centre leaves the pitch alone; clockwise goes up and counter-clockwise down, across the **SEQ OCT** range (±2 octaves in Init). With **QUANTIZE** on 12-ET each knob snaps to semitones. Or click **DICE**: step 1 stays put, and steps 2–4 land on consonant intervals from it.

2 Add an undertone

Turn up **SUB 2 LEVEL** under VCO 1. Its **SUB 2 FREQ** knob divides VCO 1's pitch: fully clockwise is unison, then 1/2 an octave down, 1/3 an octave and a fifth down, 1/4 two octaves, 1/5 two octaves and a major third. The sub follows every note the sequencer plays, in tune by construction.

3 Add a second rhythm

Light **SEQ 1** under **RHYTHM 2** (set to 1/3). Sequencer 1 now advances on every second pulse and on every third. The pattern repeats every six pulses, but the gaps between steps are uneven, and that unevenness is the groove. Press **RESET** to line both up again.

4 Bring in the second voice

Turn up **VCO 2 LEVEL**, light **SEQ 2** under **RHYTHM 4**, and turn RHYTHM 4 to 1/5. Then click Sequencer 2's **DICE**. Sequencer 2 plays VCO 2 (**OSC 2** is lit under SEQ 2 ASSIGN) on its own rhythm, and with 1/2, 1/3 and 1/5 running together the whole pattern takes 30 pulses to come back around.

5 Shape it

CUTOFF and **RESONANCE** set the tone, and **VCF EG AMT** decides how far each step opens the filter. **VCA DECAY** sets the note length: short for plucks, long to let notes ring into each other.

6 Save it

Click **SAVE** on the bottom rail and give the patch a name. Presets keep every knob, switch and patch cable, and show up in every Live set.

> **Tuning by ear** Hold **EG** until it blinks. The envelopes stay open, so you hear a steady tone. With Live stopped, **NEXT** steps through the sequence while you tune each knob. Press EG again to go back.

Find your way around



1	**Sequencers**	Two four-step sequencers. Sequencer 1 plays VCO 1, Sequencer 2 plays VCO 2. The LED shows the current step.
2	**DICE**	Keeps step 1 and rolls steps 2–4 to consonant intervals from it. MIDI-mappable as Dice Seq 1 and 2.
3	**Polyrhythm**	Four rhythm generators, each dividing the clock by 1 (clockwise) to 16. Their SEQ buttons pick which sequencer they advance.
4	**Transport**	TEMPO (free-running only), RESET, EG, NEXT, PLAY and TRIGGER.
5	**Oscillators**	VCO FREQ (four octaves up from middle C), the WAVE switch, and SUB FREQ dividing each oscillator by 1 to 16.
6	**SEQ OCT**	The step knobs' range: ±1, ±2 or ±5 octaves.
7	**SEQ ASSIGN**	What each sequencer moves: the oscillator's pitch, a sub's divisor, or any mix of the three.
8	**Mixer**	Levels for all six sources. Push several high for warm saturation.
9	**QUANTIZE**	Off, 12-ET, 8-ET, 12-JI or 8-JI, for the FREQ knobs and the steps.
10	**Filter and VCA**	CUTOFF, RESONANCE (rings on its own at maximum), VCF EG AMT (centre is none) and VOLUME.
11	**Envelopes**	Attack and decay for the filter and the VCA, shared by every source.
12	**Patchbay**	17 inputs (plain labels) and 15 outputs (white labels). Drag cables between them.
13	**Bottom rail**	Presets and SAVE, SYNC and RATE, LOGIC, FINE TUNE, DRIFT, MIDI channel and the status display.

> **Hold for more** Three buttons do something extra when held. **EG** holds the envelopes open. **QUANTIZE**, held until its LED blinks, sends unquantized CV from the SEQ jacks. **SEQ OCT**, held until its LED blinks, keeps the SEQ jacks at ±5 V.

Patch, play and record

Patching

Drag from an output (white label) to an input to add a cable. Grab a plugged input to move its cable, and right-click a jack to unplug it. **CLEAR CABLES** on the bottom rail removes them all. Three to try:

SEQ 2 → CUTOFF	Sequencer 2 plays the filter as well as VCO 2. Unassign OSC 2 to make it a filter sequence only.
CLOCK → VCA	Every clock pulse opens the VCA: a ticking pulse under everything. The Odd Rock preset uses it.
SEQ 2 → RHYTHM 1	With RHYTHM 1 at noon, each Sequencer 2 step sets how fast Rhythm 1 runs. Hold SEQ OCT until it blinks for the widest range.

Presets

Click the preset name on the bottom rail to browse eleven factory patches, then any you've saved. **Ladder Kick** uses no oscillators at all: the self-oscillating filter is the drum. An asterisk after the name means you've changed something since loading it.

Play it from a clip

MIDI notes on Fractone's track transpose both oscillators relative to C3 (MIDI note 60) and open the envelopes, so a clip can move the running pattern into a new key, or play Fractone as a monosynth with the sequencers off.

Put DICE on a controller

Click the device's **Configure** button, click a DICE button so its parameter appears, then use MIDI Map mode (**⌘M**) and press a pad. Map it as a momentary button. Draw a short pulse on the Dice Seq 1 automation lane every few bars and the phrase re-rolls as the song plays.

Record what it plays

Arming Fractone's own track records MIDI, and Fractone plays itself, so that clip stays empty. To capture the sound:

- Create an audio track and set its **Audio From** to the Fractone track, **Post FX**.
- Set its **Monitor** to **Off**, arm it, and record. Turn on **Automation Arm** to keep your knob moves as well.

Troubleshooting

Fractone isn't in the browser	Turn on **Use VST3 Plug-in System Folders** (or **Use Audio Units**) and press **Rescan**.
The display says WAITING FOR HOST	Press Live's play button. With SYNC on, Fractone only runs while Live does.
Running, but no sound	Check **VOLUME** and the mixer LEVEL knobs. If **EG** is unlit, steps don't fire the envelopes; click it, or hold it until it blinks to hold them open.
A sequencer doesn't move	It needs at least one rhythm with its **SEQ** button lit. Holding RESET, or a cable into the RESET jack, also holds step 1.
Everything pulses together	That's the design: one filter and one VCA shared by all six sources. For a drone under a pulse, use two Fractone tracks, one with EG held.
Recording captures nothing	The armed MIDI track records MIDI. Record onto an audio track with **Audio From** set to the Fractone track.
Dice Seq 1 isn't in Live's MIDI map	Click **Configure** on the device, then click that DICE button so the parameter appears.
The preset name has an asterisk	You've changed something since loading it. **SAVE** keeps the change.

Everything else, control by control, is in the README.